CCTV SECRET & ENTREPRENEURSHIP • BY ALI ANSARI

Chapters 1 & 2: Illustrated Technical Summary

Foundations of CCTV Technology, Camera Architectures, Connectors, Power Supplies & Field Tools

Chapter 1: Introduction to CCTV Technology

Closed-Circuit Television (CCTV) is a dedicated surveillance system where video cameras transmit signals to a specific set of monitors and storage devices rather than broadcasting publicly. Its core purpose is continuous monitoring, audio/video capture, and hard-disk recording for security, safety, and operational oversight.

Key Applications of CCTV

Security & Home Surveillance

Deters crime, protects residential and government properties, and provides recorded evidence for investigations.

Commercial & Process Monitoring

Prevents workplace theft, monitors employee safety, and oversees factory manufacturing lines in real time.

Traffic Monitoring

Tracks traffic flow at intersections, improves road safety, and enables rapid response to road incidents.

Public Spaces

Deployed in malls, parks, and transport hubs to support crowd management and public emergency response.

Evolution of CCTV Technology (4 Phases)

Phase	Era & Hardware	Key Technological Leap
Phase I	1942 (Germany)	Used to observe A4 rocket launches; cameras wired directly to live monitors with zero recording capability.
Phase II	1960s – 1970s	Introduced switch boxes, solid-state cameras, split-screen multiplexers, and VCR tape recording for judicial evidence.
Phase III	DVR & IP Era	Digitized video onto hard drives (DVR), enabling simultaneous live view and playback plus remote IP access over the internet.
Phase IV	Modern PTZ & Smart	Pan-Tilt-Zoom cameras with multi-lens night vision, real-time two-way audio, and facial recognition capabilities.

Types of CCTV Systems & Camera Comparison



HD Dome & Bullet Cameras

Dome: Indoor use. **Bullet:** Outdoor use. Cost-effective (VGA to 5 MP), ~5 yrs durability, easy setup.



IP CCTV Cameras

Digital network cameras (2.0 to 8 MP) offering superior clarity, remote internet access, and 8+ yrs durability.



C-Mount Cameras

Standardized screw-thread interface supporting interchangeable lenses for wide-angle or long-distance zoom.



PTZ (Pan-Tilt-Zoom) Cameras

Remotely move left/right, up/down, and zoom in/out with preset positions and automatic motion tracking.

System Infrastructure Workflows

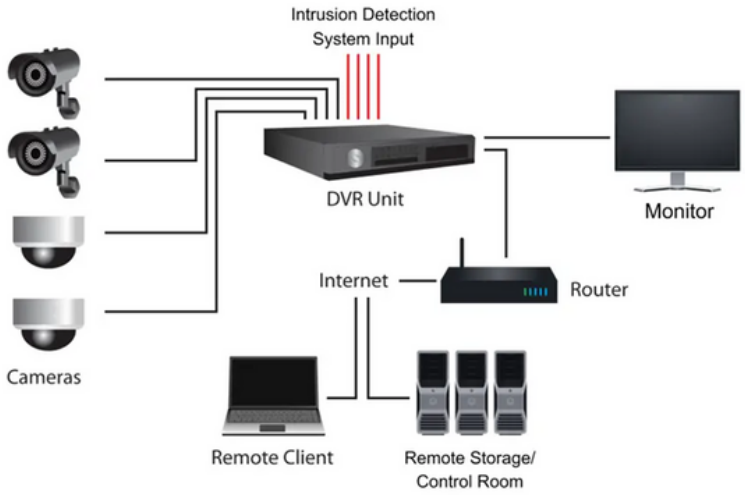


Figure 1.1: HD Camera Infrastructure Workflow — Cameras connect to the DVR Unit, which outputs to a local Monitor and connects via a Router/Internet to Remote Clients and Control Room Storage.

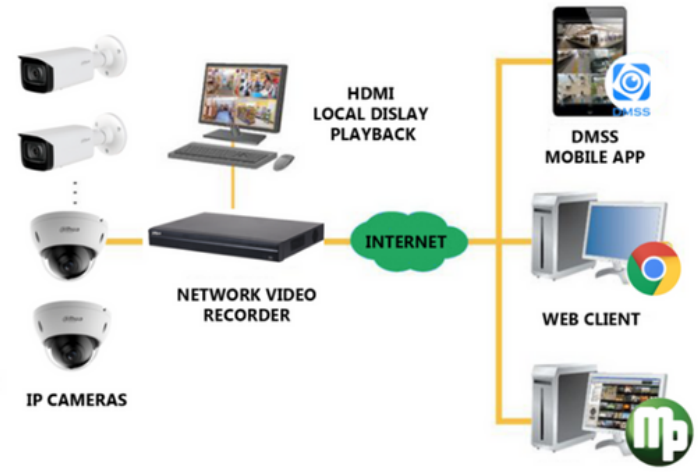


Figure 1.2: IP Camera Infrastructure Workflow — IP Cameras send digital streams to the Network Video Recorder (NVR) for HDMI local playback and remote viewing via the DMSS Mobile App and Web Client.

Chapter 2: Components & Tools

Every reliable surveillance installation depends on matching the right connectors, power supplies, and field tools to the camera architecture. Chapter 2 breaks down each component so technicians can build clean, fault-free systems.

Core Connectors & Signal Components



BNC Connector

Transmits analog/HD video over 3+1 coaxial cable between HD cameras and DVR/XVR units.



RJ45 Connector

Terminates Cat6 Ethernet cables to link IP cameras with NVR devices and PoE switches.



Video Balun (Twisted Pair)

Enables HD camera video transmission over Cat6 twisted-pair cable across longer distances with minimal interference.



DC & Audio Connectors

DC jacks deliver 12V power; RCA/3.5mm audio connectors integrate microphones and speakers.

Standard 3-Wire Color Coding (Coaxial Setup)

In 3-wire CCTV cabling, **Yellow** carries the Audio signal, **Red** carries Power (+), and **Black** carries Power (–). Keeping color polarity consistent prevents short circuits and speeds up maintenance.

Power Supply & Accessories



SMPS (Switched Mode Power Supply)

Converts AC mains to regulated 12V DC for centralized multi-camera power, protecting against voltage surges.



PoE (Power over Ethernet)

Delivers both data and electrical power over a single Cat6 cable, eliminating separate power adapters for IP cameras.

Essential Accessories: Monitor mounts/brackets, enclosed DVR/NVR security racks, wireless mice for recorder navigation, Wi-Fi dongles/routers for remote internet access, and 12V 1A individual power adapters for distributed setups.

Essential Tools for CCTV Technicians



Digital Multimeter

Measures voltage (V), current (A), and resistance (Ω) to verify power health and diagnose wiring faults.



LAN Cable Tester

Verifies Cat6 Ethernet pinouts, detects open/shorted pairs, and confirms network continuity before mounting.



LAN Crimping Tool

Terminates Cat6 cables with RJ45 connectors cleanly, minimizing signal loss and loose connections.



Mounting & Finishing Tools

Drill machine & matching bits, screwdrivers, pliers, hammer, wire cutter, measuring tape, black insulation tape, cleaning brush, and tool bag.